Adaptive Staircase
PSY 310: Lab in Psychology
Lab Report

Date: 8th August 2026
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Github Link

Introduction

Adaptive staircase methods are commonly used in psychophysics to measure sensory thresholds by changing the intensity of a stimulus according to the participant's responses. Instead of presenting stimuli at fixed intensities, the staircase procedure makes the task easier or harder depending on whether the participant responds correctly. This allows the experiment to efficiently estimate the level at which a person can reliably detect a difference in a stimulus (Cornsweet, 1962; Leek, 2001). Adaptive procedures are widely used in areas such as vision and hearing research because they can measure thresholds without requiring a large number of trials at every possible stimulus level (Leek, 2001). In the present experiment, an adaptive staircase was used to measure the participant's ability to discriminate the direction of a tilted grating. The experiment followed a one-up, three-down procedure, where the tilt intensity was adjusted according to the participant's responses. PsychoPy was used to present the stimuli and record the responses (Peirce et al., 2019).

Method

Participant: The participant was a 21-year-old undergraduate student at Ahmedabad University. The participant was informed about the general purpose of the experiment, which was to identify the direction of the tilt presented in the stimulus. Consent was obtained before the experiment was conducted.

Materials and Procedure

The experiment was created and run using PsychoPy Builder version 2026.2.3. The experiment was presented in a window of 1000 × 600 pixels. The display used a black background and the actual frame rate recorded during the experiment was approximately 60.13 Hz.

The main stimulus was a sinusoidal grating with a Gaussian mask. The grating had a size of 0.3 × 0.3, a spatial frequency of 8, and a contrast of 0.5. The stimulus was presented at different levels of tilt. On each trial, the direction was randomly selected so that the grating could be tilted either to the left or to the right. The correct response was therefore either the left or right arrow key.

A white cross-shaped fixation stimulus was also presented before the grating. The fixation stimulus was shown for 500 milliseconds, after which the grating appeared. The grating remained on the screen for 300 milliseconds.

Procedure

The experiment consisted of 100 trials. The adaptive staircase started with a tilt intensity of 10°. The minimum possible intensity was 1° and the maximum was 20°. The staircase used step sizes of 2°, 1°, 1°, and 0.5°, with the step size becoming smaller as the staircase progressed. A one-up, three-down procedure was used, with one incorrect response increasing the intensity and three consecutive correct responses decreasing it.

At the beginning of each trial, a fixation cross was presented for 500 milliseconds. This was followed by the sinusoidal grating for 300 milliseconds. The grating was randomly tilted either to the left or the right. The participant had to identify its direction by pressing the left or right arrow key. The response was recorded by PsychoPy, along with response time and whether the response was correct.

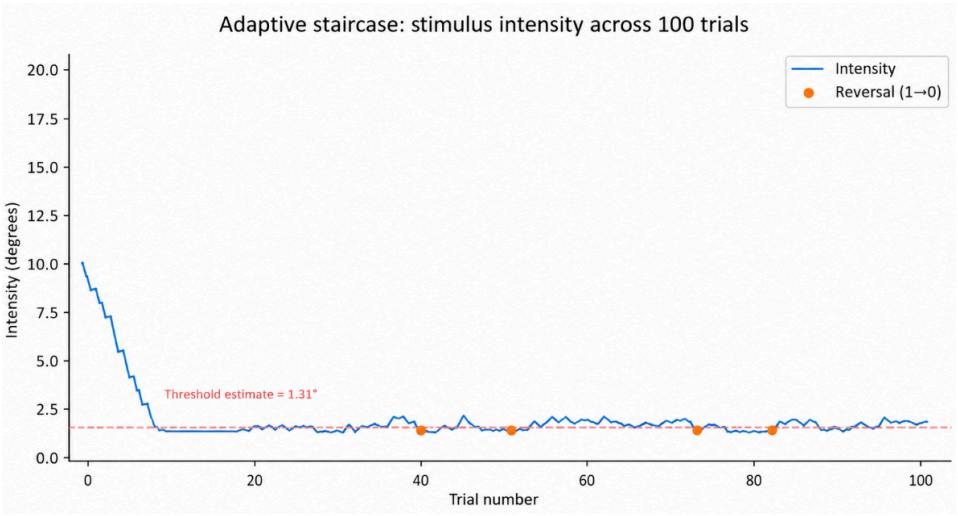
After each response, the staircase adjusted the tilt intensity according to the participant's accuracy. The data from all 100 trials were saved by PsychoPy in an Excel-compatible

data file. The recorded data included the stimulus intensity, direction of tilt, response, accuracy, and reaction time.

Reversal	Trial	Intensity
1	52	1
2	76	1.5
3	79	1
4	82	1.5
5	85	1
Average		1.2

Results

The measured difference threshold is 1.2 degrees. The participant completed all 100 trials. Out of the 100 responses, 96 were correct and 4 were incorrect, giving an overall accuracy of 96%. The staircase quickly moved from the higher intensities towards the lower intensities because the participant was able to correctly identify the direction of the grating. The overall accuracy of 96% also indicates that the participant was highly consistent in identifying the direction of the grating. The staircase data show that this high accuracy was maintained even when the tilt was reduced to very small values.



Discussion

The results show that the participant had good sensitivity to differences in orientation, as shown by the high accuracy of 96% and the low estimated threshold of 1.31°. The staircase method successfully reduced the difficulty of the stimulus until it reached the participant's approximate discrimination threshold. However, since the experiment was conducted with only one participant, the results cannot be generalised to a larger population. The participant's attention, familiarity with the task, and individual visual sensitivity may also have affected the results.

References

Cornsweet, T. N. (1962). The staircase-method in psychophysics. *The American Journal of Psychology, 75*(3), 485–491.

Signal Detection

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Introduction

Signal detection theory is used to understand how people distinguish between a signal and background noise, while also accounting for the possibility of errors. Performance can be influenced not only by a person's ability to detect a signal, but also by their response criterion, or how willing they are to report that a signal is present. Signal detection theory therefore separates sensitivity from response bias, with measures such as hit rate, false alarm rate, and d′ being used to assess performance. In an orientation discrimination task, participants have to detect whether a stimulus contains a particular orientation or tilt and respond accordingly. The present experiment examined the participant's ability to discriminate between different orientations and used signal detection measures to assess sensitivity and response criterion. The experiment was conducted using PsychoPy, which allows visual stimuli and participant responses to be presented and recorded systematically (Peirce et al., 2019).

Method

Participant

The participant was a 21-year-old undergraduate student at Ahmedabad University. The participant was informed about the task and the required responses before the experiment began and gave consent to participate.

Materials

The experiment was created using PsychoPy Builder version 2026.2.3. It used a fullscreen display with a window resolution of 1024 × 768 pixels. The experiment measured the actual frame rate of the display during setup.
The visual stimulus was a sinusoidal grating with a Gaussian mask. The grating had a size of 0.2 × 0.2, a spatial frequency of 5, and a contrast of 0.3.

A fixation stimulus was presented at the beginning of each trial. The fixation was displayed for 1 second. The grating was then presented for 300 milliseconds.

Procedure

The experiment consisted of 100 trials. Each trial began with a fixation stimulus presented for 1 second, followed by a sinusoidal grating with a Gaussian mask for 300 ms. Different tilt values were presented across trials, including zero-tilt and tilted stimuli. The participant responded using the up or down arrow keys to indicate the stimulus condition. The experiment was run using PsychoPy 2026.2.3, and the participant's responses and accuracy were recorded for each trial.

Results

Hits	54
Misses	3

False Alarm	10
Correct Rejection	32
False Alarm Rate	4.2
Hit Rate	0.94

Using standard signal detection theory:

$d' = Z(H) - Z(FA)$
$d' = Z(0.947) - Z(0.238)$
$d' = 2.33$

Criterion: $c = -0.5[Z(H) + Z(FA)]$
$c = -0.46$

The participant achieved an overall accuracy of 86%, with 54 hits, 3 misses, 10 false alarms, and 32 correct rejections. The hit rate was 94.7%, while the false alarm rate was 23.8%. The participant's d′ was approximately 2.33, indicating reasonably good sensitivity in distinguishing between the two stimulus conditions.

Discussion

The results suggest that the participant performed well on the orientation discrimination task. The high hit rate and d′ value indicate good sensitivity to the stimulus, although the false alarm rate shows that some signal-absent trials were incorrectly identified as containing the signal. Since the experiment was conducted with only one participant, the findings cannot be generalised to a larger population. Other factors such as attention and response strategy may also have influenced performance.